INTRODUCTION

Prior lessons priced fixed-income instruments by discounting all future cash flows using a single interest rate, such as the yield-to-maturity or a market reference rate (MRR) plus a discount margin. The next three lessons relax this assumption by introducing the term structure of interest rates, or the fact that interest rates vary with time-to-maturity. The ideal data to use for term structure analysis are default-risk-free zero-coupon bonds, known as spot rates or the spot curve. Since these are generally not directly observable, various estimation techniques are used. The spot curve is used to derive two other important yield curves: the par curve and the forward curve. A par curve involves bond yields for hypothetical benchmark securities priced at par, while the forward curve involves rates for interest periods starting in the future. All three of these curves are fundamental to fixed-income analysis and other applications because they represent default-risk-free rates of return for time periods that start today and in the future. We show the pricing of bonds using these different rates and establish their relationships.

Learning Module Overview

- Spot rates are market discount rates on default-risk-free zero-coupon bonds, sometimes referred to as zero rates. By using a sequence of spot rates in calculating bond prices, a no-arbitrage bond price is obtained.
- A par rate is the market discount rate for a specific maturity that would result in a bond priced at par. A par rate is derived from the spot rates up to and including the maturity date.
- Implied forward rates are calculated using spot rates and can be interpreted as an incremental, or marginal, return for extending the time-to-maturity for an additional time period. As such, they reflect a breakeven reinvestment rate.
- Since par and forward rates can be derived from spot rates, the shape of the spot curve is closely related to the shape of the par and forward curves.
- In upward-sloping term structures, par rates will be lower than their corresponding spot rates and forward rates will be greater than spot rates. In downward-sloping term structures, par rates will be greater than spot rates and forward rates will be lower than spot rates.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. Calculate the price and yield-to-maturity (YTM) of a 1% coupon, three-year bond with par value of 100 given the following spot rates:

1-year	1.0%
2-year	1.5%
3-year	2.0%

Hide Solution

$$PV = \frac{PMT}{(1+Z_1)^1} + \frac{PMT}{(1+Z_2)^2} + \cdots + \frac{PMT+FV}{(1+Z_N)^N}$$

$$PV = \frac{1}{(1+0.01)^1} + \frac{1}{(1+0.015)^2} + \frac{101}{(1+0.02)^3}$$

$PV = 97.135$

The YTM is 1.993%:

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \cdots + \frac{PMT+FV}{(1+r)^N}$$

$$97.135 = \frac{1}{(1+r)^1} + \frac{1}{(1+r)^2} + \frac{101}{(1+r)^3}$$

$r = 0.01993 = 1.993\%$

2. Define what a par rate represents in the context of bond pricing.

Hide Solution

Solution:

The par rate is the yield-to-maturity that makes the present value of a bond's cash flows (i.e., its price) equal to its par or face value.

3. Calculate an implied one-year forward rate two years from now (i.e., 2y1y) given the following spot rates, assuming annual compounding:

1-year	1.0%
2-year	1.5%
3-year	2.0%

Hide Solution

Solution:

$(1 + Z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + Z_B)^B$.

$(1 + 0.015)^2 \times (1 + IFR_{2,1})^1 = (1 + 0.02)^3$.

$IFR_{2,1} = 3.01\%$.

4. Explain the economic meaning of the forward rate result calculated in Question 3.

Hide Solution

Solution:

The one-year forward rate two years from now (2y1y) of 3.01% implies that investors expect the one-year interest rate to be 3.01% at the end of two years. This reflects the rate at which reinvestment would yield the same results over a three-year horizon regardless of whether the investor initially bought a two-year bond versus a three-year bond.

5. Contrast spot and forward rates.

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Solution:

Spot rates reflect borrowing and investing rates for funds immediately (or in the next few days) over the specified tenor, while forward rates reflect rates

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The Term Structure of Interest Rates: Spot, Par, and Forward Curves

spot rates when the spot rate curve is downward sloping.

A. True
B. False

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Solution:

False. When the spot curve is downward sloping, the one-year forward rate curve will lie below the spot rate curve. Declining spot rates for longer maturities imply that investors should expect lower rates of reinvestment in future years, and the forward curve reflects these expectations.

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AS

There is an error in question 3 which is corrected in question 4, however the whole calculation in question 3 is wrong. The 2y1y rate is equal to 3.01% (not 2.50%).

Created 7 months ago by Adam Stasinski

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NK

Is the solution for Q3 wrong. It should be (1.020)3=(1.015)2·(1+f2,1) = 3.01%

Created 4 months ago by Nishant Kumar Pathak

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